

SRI LANKA TECHNOLOGICAL CAMPUS

Faculty of Computing and IT



CC3356 – Natural Language Processing

Assignment Project Report

Assignment Title: Cross-Platform Spam Detector

Group Number: Group 17

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1. Introduction

1.1 Problem Background

Spam messages are one of the biggest concerns in digital communication in modern times. There are plenty of unsolicited and dangerous messages sent every day through different channels including SMS, Email, and YouTube comments. Such messages cause problems with cluttering user's mailbox, wasting time, performing various scams, and stealing users' personal information. Checking every message manually is inefficient and takes a lot of time. Hence, a mechanism is required to detect such messages in a timely manner.

1.2 Objectives

- To construct a single dataset by combining text messages from SMS, Email, and YouTube comments.
- To develop an end-to-end NLP pipeline for text data preprocessing.
- To train and test 3 machine learning models (Logistic Regression, SVM, Random Forest) and 3 deep learning models (LSTM, GRU, BERT).
- To develop a user-friendly web application which is capable of detecting spam messages in real-time and highlighting malicious URLs and keywords.

1.3 Importance of the Solution

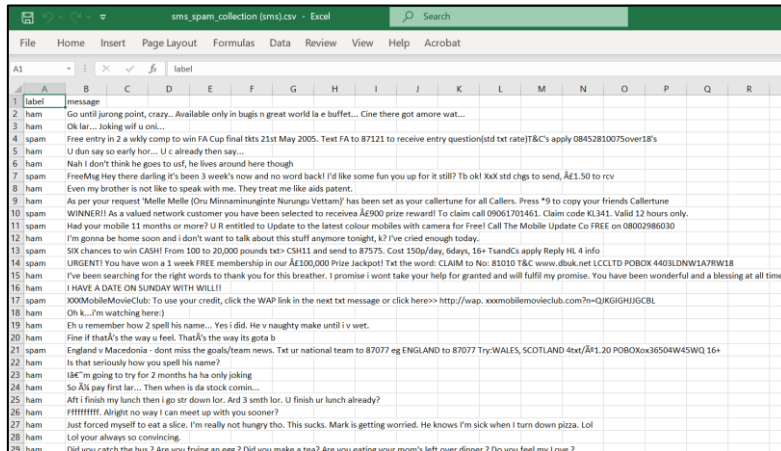
This project will be useful to ordinary users, students, content creators, and businesses in order to save their time and stay secure online.

2. Dataset Description

- **Dataset Name:** Combined Cross-Platform Spam Dataset
- **Dataset Source (URL):**
 - Source 1 (SMS): <https://www.kaggle.com/datasets/hamnawaseem112222222/sms-spam-collection-5572-labeled-sms-messages>
 - Source 2 (Email): <https://www.kaggle.com/datasets/venkateshch22384/spam-and-ham-email-dataset>
 - Source 3 (YouTube): <https://www.kaggle.com/datasets/fatihblg/youtube-spam-collection>
- **Number of Records:** 11,911 total combined rows (SMS: 5,572 + Email: 4,845 + YouTube: 1,494)

2.1 Sample Records (Excel Dataset View)

- SMS dataset



A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	label	message															
2	ham	Go until jurong point, crazy.. Available only in bugis n great world la e buffet... Cine there got amore wat...															
3	ham	Ok lar... Joking wif u oni...															
4	spam	Free entry in 2 a wily comp to win FA Cup final tkts 21st May 2005. Text FA to 87123 to receive entry question[nt txt rate]T&C's apply 08452810070over18's															
5	ham	U dun say so early hor... U c already then say...															
6	ham	Nah I don't think he goes to usf, he lives around here though															
7	spam	FreeMsg Hey there darling it's been 3 week's now and no word back! I'd like some fun you up for it still? Tb ok! Xxxt std chgs to send, A£1.50 to rcv															
8	ham	Even my brother is not like to speak with me. They treat me like aids patient.															
9	ham	As per your request 'Melle Melle (Ovu Misanjunginte Nuringu Vettaru)' has been set as your calltune for all Callers. Press *9 to copy your friends Calltune															
10	spam	WINNER!! As a valued network customer you have been selected to receive a £900 prize reward! To claim call 09061701461. Claim code KL341. Valid 12 hours only.															
11	spam	Had your mobile 11 months or more? U R entitled to Update to the latest colour mobiles with camera for Free! Call The Mobile Update Co FREE on 08002986030															
12	ham	I'm gonna be home soon and i don't want to talk about this stuff anymore tonight, k? I've cried enough today.															
13	spam	SIX chances to win CASH! From 100 to 20,000 pounds! Txt CS111 and send to 87575. Cost 150p/day, 6days, 15+ TandCs apply Reply M. 4 info															
14	spam	URGENT! You have won a 1 week FREE membership in our £100,000 Prize Jackpot! Txt the word: CLAIM to No: 81010 T&C: www.dsk.net LCCLTD POBOX 4403LDNW1A7R9W18															
15	ham	I've been searching for the right words to thank you for this breather. I promise i wont take your help for granted and will fulfil my promise. You have been wonderful and a blessing at all times.															
16	ham	I HAVE A DATE ON SUNDAY WITH WILL!!															
17	spam	XXXMobileMovieClub: To use your credit, click the WAP link in the next txt message or click here>> http://wap.xxxmobilemovieclub.com/?n=QJGKHUJGCB															
18	ham	Oh k.. I'm watching here)															
19	ham	Eh u remember how 2 spell his name... Yes i did. He v naughty make until i wct.															
20	ham	Fine if that's the way u feel. That's the way i gotta b															
21	spam	England v Macedonia - dont miss the goals/team news. Txt ur national team to 87077 eg ENGLAND to 87077 Try/WALES, SCOTLAND 4txt/A1. 20 POBOX36504W45WQ 16+															
22	ham	Is that seriously how you spell his name?															
23	ham	l8c"m going to try for 2 months ha ha only joking															
24	ham	So A\$ pay first lar... Then when is da stock comin...															
25	ham	Alti I finish my lunch then i go str down lor. Ard 3 smth lor. U finish ur lunch already?															
26	ham	FFFFFFF. Alright no way I can meet up with you sooner?															
27	ham	Just forced myself to eat a slice. I'm really not hungry tho. This sucks. Mark is getting worried. He knows I'm sick when I turn down pizza. Lol															
28	ham	Lol you're always so convincing.															
29	ham	Did you catch the bus? Are you trying an ope? Did you make a tea? Did you make a tea? Did you eatine your mom's left over dinner? Do you feel my Love?															

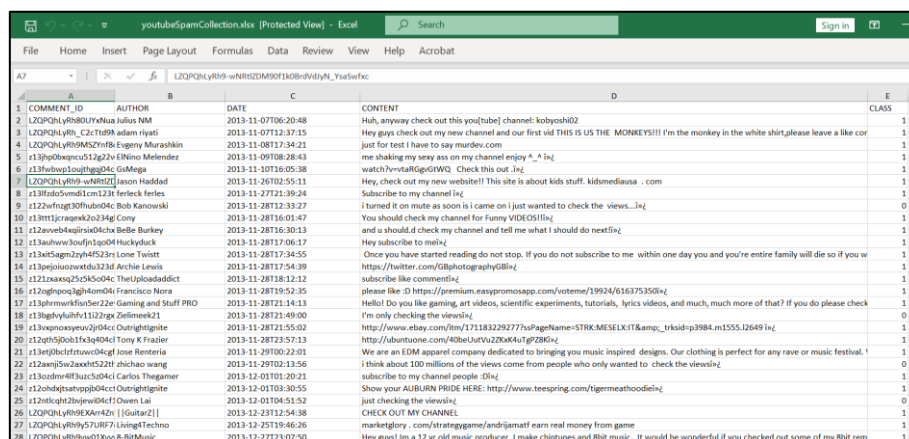
Figure 2.1: Screenshot of raw SMS Spam Collection dataset in MS Excel.

- Email dataset

spamham(email).csv - Excel		Search	Sign in					
File	Home	Insert	Page Layout	Formulas	Data Review	View	Help	Account
A1								
Text								
A								
B								
Class								
Label								
1 Supply Quality China's EXCLUSIVE dimensions at Unbeatable Price.Dear Sir, We are pleased to inform you as one of China's largest export & import sto-ne company-Wanlstone Group, The Group its subsidiaries specialize in the e								
over. Sell me know. The								
2 Dear Friend,Greetings to you! wish to account you with a request that would be of immense benefit to both of us. Being an executor of wills, it is possible that we may be tempted to make fortune out of our client's situations, wh								
3 MIL CHUNG PUHANG SENG BANK LTD.DES VOEUX RD. BRANCH,CENTRAL HONG KONG,HONG KONG. Let me start by introducing myself. I am Mr.Cheung Pui director of operations of the Hang Seng Bank Ltd. I have an urgent t								
4 Not a surprising assessment from Embassy.								
5 Monica -Huma Abedin<clumsy@state.gov>Tuesday June 29 2010 6:01 AMHuma@state.gov: Wife is already in locked for tonite. I am seeing her right before actually.								
6 Pis print.H<chrod7@clintonemail.com>Thursday October 8 2009 8:01 PMJilony.C@state.gov: WHI - powder coatingB6								
7 Dear Tom-M<chrod7@clintonemail.com>Friday December 11 2009 5:41 PMCould we schedule a call?I have several matters I'd like to discuss w you this weekend and was hoping we can schedule a call. I'm free tonightafter 10;								
8 Greetings from barrister Robert Williams-2CDear friend-2C I know that my letter will come to you as a surprise-2C based on the fact that we have not been in contact for the first time eit-her in person or by correspondent-2E								
9 PY. Thanks again for signing the book - and I do hope you get royalties from Mongol! Also thought you would like to see the attached letter to President Clinton.								
10 Pis putRLEASE IN PARTB6H<chrod7@clintonemail.com>Friday July 31 2009 3:25 PMJilony.C@state.gov: Russov@state.gov: Bravalon call list and do letter to her wishing her well etc. B6								
11 Not sure about SRAP. I'll forward it on. Hope the trip is going well.Im								
12 SOLICITING FOR A BUSINESS VENTURE AND PARTNERSHIP DEAR SIR AND THE SON OF THE LATE PRESIDENT OF THE FEDERAL REPUBLIC OF ZAMBIE, «PRESIDENT MOBUTU SESE SEKO (NOW REPUBLIC OF CONGO UNDER THE LEAD								
13 Dear Sir/Madam,Compliments of the season. It is indeed my pleasure to write to you thisletter, which i believe you will welcome it. Mr Edward Govou, the manager-creditand foreign bills of Eco Bank du Benin-Rdp.I am writing								
14 Dear Bill and Tom,I'm sorry that I cannot be on the Hill today as we had long planned but very grateful that you both will be. TheDepartment and I appreciate your leadership everyday,so I'll be nursing my cracked head and cheeri								
15 I am Dr. Usman al, I am a former specialAdviser on Petroleum matters to the late Head of StateOf Nigeria Gen. Sani Abacha. Because of my strategicPosition in the former Government and also being aClose confidant of the late								
16 The letter is good to go as we. HRC.								
17 Yes - Stephens is in DC all day we can work with Ops on the call. I don't think we need to push the Angolans more than15 minutes so should be fine. Should I move forward and adjust?								
18 Yes Iona has it on list								
19 Required information:1) Your full name and address2) Your private phone, mobile and fax number where you can always be reach. As soon as your information is been received by my attorney, he will also issue a Letter of Autho								
20 Burns this is going to happen. We are committed to network and other docs like her. Lyn heard rumors about acemepetive hospital etc. We will work this out JIBC did the event at her hospital after all.								
21 DEAR SIR/MADAM,MY NAME IS ENGINEER. TONY EDEM THE DIRECTOR GENERAL WITH THE MINISTRY OF WORK-IRS AND HOUSING HERE IN NIGERIA, I WISH TO EXTEND THIS BUSINESS RELATIONSHIP-TO YOU WITH THE HOPE								
22 Madame Secretary:Thank you for reaching out to Secretary Solis and convincing her to join the Verification Commission. She and RicardoLagos will make for a very high profile and effective international component of the Com								
23 Fv								
24 Dear Sir or Madam:We are keenly interested in setting up a new factory to produce 120,000 ERW oil casings per year to meet the big market of Shengli Oilfield. Since we lack some starting-up capital, we are seeking for investor								
25 8:00 am Call hr / israel PM Netanyahu8:15 am Private Residence8:15 am DEPART Private Residence8:25 am ARRIVE State Department8:25 am PRESIDENTIAL DAILY BRIEFING8:30 am Secretary's Office								
26 AbdelHuma Abedin<clumsy@state.gov>Wednesday July 15 2009 1:44 PMRe: ACTWILL be off campus at meetinfant you should definitely come I think they have found some good things.								

Figure 2.2: Screenshot of raw Email Spam dataset in MS Excel.

- Youtube dataset



A	B	C	D	E
1	COMMENT_ID	AUTHOR	DATE	CONTENT
2	1ZQPQHLYh8H0UYkxJufus NM	2013-11-07T06:20:48	Huh, anyway check out this you[tube] channel: kobayosh02	1
3	1ZQPQHLYh8H0UYkxJufus NM	2013-11-07T12:37:15	Hey guys check out my new channel and our first vid THIS IS US THE MONKEYS!!!! I'm the monkey in the white shirt, please leave a like cor	1
4	1ZQPQHLYh8H0UYkxJufus NM	2013-11-08T17:34:21	just for test i have to say murder.com	1
5	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-09T08:28:43	me shaking my sexy ass on my channel enjoy ^_ ^_!az	1
6	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-10T16:05:38	watch?v=vtaR6pGfWQW Check this out...!az	1
7	1ZQPQHLYh8H0UYkxJufus NM	2013-11-26T02:55:11	Hey, check out my new website!! This site is about kids stuff. kidmediausa . com	1
8	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-27T21:39:24	Subscribe to my channel !az	1
9	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-28T12:32:27	I turned it on mate as soon as i came on i just wanted to check the views...!az	1
10	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-28T16:01:47	You should check my channel for Funny VIDEOS!az	1
11	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-28T16:30:13	and u should.d check my channel and tell me what i should do next!az	1
12	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-28T17:06:17	Hey subscribe to me!az	1
13	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-28T17:34:55	Once you have started reading do not stop. If you do not subscribe to me within one day you and you're entire family will die so if you w	1
14	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-28T17:54:39	https://twitter.com/GlaphographyGB!az	1
15	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-28T18:12:12	subscribe like comment!az	1
16	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-28T19:52:35	please like -D https://premium.easypromosapp.com/voteme/19924/616375350!az	1
17	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-28T21:14:13	Hello! Do you like gaming, art videos, scientific experiments, tutorials, lyrics videos, and much, much more of that? If you do please check	1
18	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-28T21:49:00	I'm only checking the views!az	1
19	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-28T21:55:02	http://www.ebay.com/itm/171183229277?usPageName=STRK:MESELX:IT&_trksid=p3984.m1555.12649!az	1
20	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-28T23:57:13	http://ubuntune.com/40beUutVuzZ0x04uTgZBK!az	1
21	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-29T00:22:01	We are an EDM apparel company dedicated to bringing you music inspired designs. Our clothing is perfect for any rave or music festival. I	1
22	13j9ebwgvcc13j2vUzElnio Melendez	2013-11-29T02:15:56	i think about 100 millions of the views come from people who only wanted to check the views!az	1
23	13j9ebwgvcc13j2vUzElnio Melendez	2013-12-01T01:20:21	subscribe to my channel people !D!az	1
24	13j9ebwgvcc13j2vUzElnio Melendez	2013-12-01T03:30:55	Show your AUBURN PRIDE HERE: http://www.teespring.com/vigametheadhood!az	1
25	13j9ebwgvcc13j2vUzElnio Melendez	2013-12-01T04:51:52	just checking the views!az	0
26	13j9ebwgvcc13j2vUzElnio Melendez	2013-12-23T12:54:38	CHECK OUT MY CHANNEL	1
27	13j9ebwgvcc13j2vUzElnio Melendez	2013-12-25T19:46:26	marketglory . com/strategymag/andriamaff earn real money from game	1
28	13j9ebwgvcc13j2vUzElnio Melendez	2013-12-27T23:07:50	Hey guys! Im a 12 yr old music producer. I make chip tunes and Bit music. It would be wonderful if you checked out some of my Bit rem	1

Figure 2.3: Screenshot of raw YouTube Spam Collection dataset in MS Excel.

- Combined dataset

A	B	C
text	label	source
supply quality china exclusive dimension unbeatable price dear sir pleased inform one china largest	1	Email
sidlet know thx	0	Email
dear friend greeting wish accost request would immense benefit u executor will possible may temp	1	Email
mr chung puhang seng bank ltd de voeux rd branch central hong kong hong kong let start introduc	1	Email
surprising assessment embassy	0	Email
monica huma abedin tuesday june hanley state gov hre already locked tonite seeing right actuall	0	Email
pi print thursday october pm jilolyt state gov fw whi powder coatingb	0	Email
dear tom friday december pmcould schedule call several matter like discuss weekend hoping schedu	0	Email
greeting barrister robert williams cdear friend know letter come surprise ased fact contact first tim	1	Email
fy thanks signing book hope get royalty mongolia also thought would like tosee attached letter pre	0	Email
pls putrelease partb friday july pm jilolyt state gov russorv state gov fw bravalon call list letter wis	0	Email
sure strap forward hope trip going well jm	0	Email
soliciting business venture partnership dear sir son late president federal republic zaire president mc	1	Email
dear sir madam compliment season indeed pleasure write thisletter believe welcome mr edward gi	1	Email
dear bill toni sorry cannot hill today long planned grateful thedepartment appreciate leadership ev	0	Email
dr usman ali former specialadviser petroleum matter late head stateof nigeria gen sani abacha stra	1	Email
letter good go hrc	0	Email
yes stephen dc day work ops call think need push angolan minute fine move forward adjust	0	Email
yes lona list	0	Email
required information full name address private phone mobile fax number always reach soon inform	1	Email
burn going happen committed herwork doc like lyn heard rumor acompetitive hospital etc work hri	0	Email
dear sir madam name engineer tony edem director general ministry wo rks housing nigeria wish ext	1	Email
madame secretary thank reaching secretary solis convincing join verification commission ricardolaj	0	Email
fyi	0	Email
dear sir madam keenly interested setting new factory produce erw oil casing per year meet big mai	1	Email
call to israeli pm netanyahu private residence depart private residenceen route state department ar	0	Email
abedin huma wednesday july pmre artwill campus meetingbut definately come think found good th	0	Email

Figure 2.4: Screenshot of final combined cross-platform spam dataset in MS Excel.

2.2 Sample Data Overview Table

Table 2.1: Sample Overview of the Combined Cross-Platform Spam Dataset

Text Sample	Platform	Class Label	Numeric Label
Free Entry! Reply to this message now to win tickets to the FA Cup Final. T&C apply.	SMS	Spam	1
Are you free today? Let us meet up for lunch around 1:00 PM near the cafeteria.	SMS	Ham	0
Congratulations! You have been selected to win a cash prize of \$10,000. Click here to claim your reward.	Email	Spam	1
Dear team, please find attached the updated project schedule for our meeting tomorrow afternoon.	Email	Ham	0
Hey guys, check out my channel for awesome music videos! Please like and subscribe!	YouTube	Spam	1
This explanation was very helpful and easy to understand. Thank you for sharing this video!	YouTube	Ham	0

Note on Text Representation: "In the raw dataset, text features are preprocessed using techniques like stop-word removal and lowercasing. For clarity and ease of reading in this report, the sample table above presents normalized, human-readable equivalents that clearly illustrate the pattern differences between Spam and Ham messages across the three platforms."

2.3 Dataset Challenges

- **Imbalanced Classes:** The classes in the dataset are imbalanced in terms of the number of records in each class, since normal records (Ham) are more than spam records (SMS dataset contains 87% Hams and just 13% Spams). It might lead to model biasness towards a particular class.

- **Noisy Text:** The text in the data consists of high-level noise like use of internet slang language ("u", "plz"), spelling errors, emoticons, character manipulation for evasion of filtering process ("m0ney", "FR3E").
- **Ethical Concerns:** Considering that the source of our data is genuine users' message and comments on social media platforms, there is a slight level of privacy concern. It should always be ensured that no data leaks are done to protect personal information.

3. NLP Pipeline

3.1 Pipeline Approach

To ensure specialised platform cleaning and strong cross-platform performance, our methodology followed a two-stage approach:

- **Platform-Specific Preprocessing:** Each group member cleaned, normalised and analysed a specific platform's dataset (SMS, Email or YouTube) with tailored rules for that text structure.
- **Combined Dataset Training:** After individual preprocessing steps, the three datasets were merged to create a single Combined Cross-Platform Spam Dataset with 11,818 records. To make the models capable of detecting spam on any platform, all the machine learning and deep learning models (Logistic Regression, SVM, Random Forest, LSTM, GRU, and BERT) were trained and evaluated on this combined dataset.

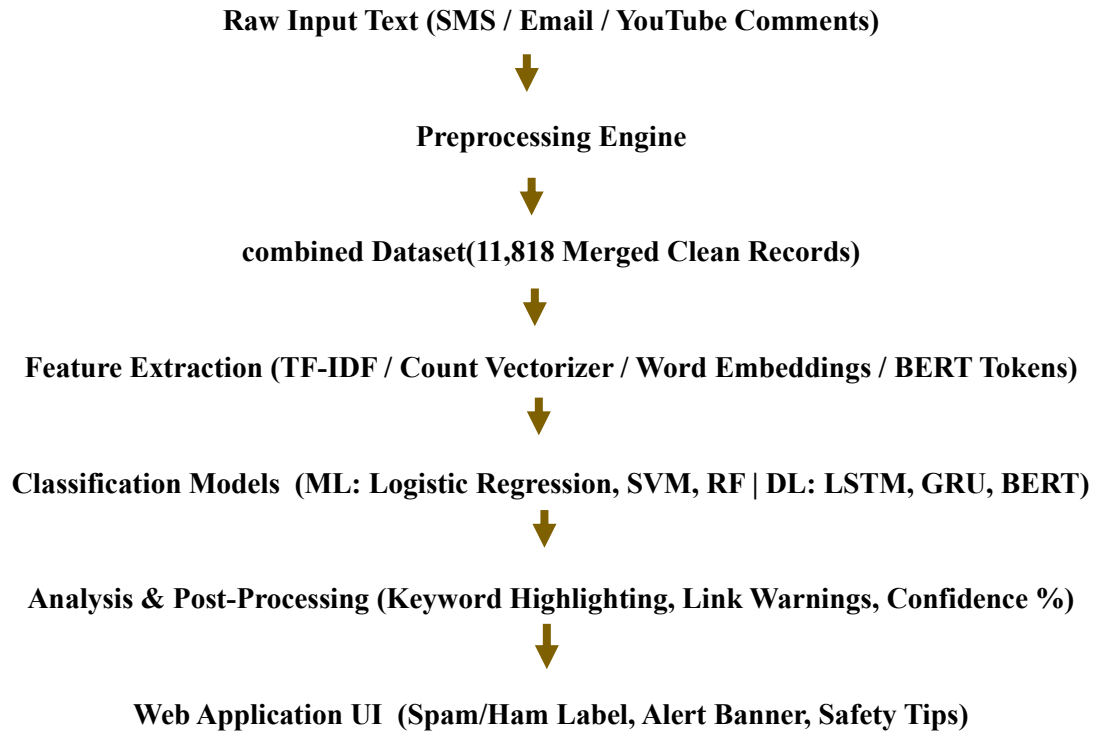
Table 3.1: Individual Group Member Responsibilities, Preprocessing Strategies, and Model Implementations

Group Member	Preprocessing Responsibility (Individual)	Models Trained on Combined Dataset	Feature Extraction Methods
P.D.M. Sathsarani	SMS Dataset • Lowercasing & noise removal • NLTK Stop-word filtering • Porter Stemming	• Logistic Regression • LSTM (Deep Learning)	• TF-IDF Vectorizer (for Logistic Regression) • Keras Tokenizer & Sequences (for LSTM)
M.K.D.S. Amarasinghe	Email Dataset • HTML tag & header stripping • Regex normalization • WordNet Lemmatization	• Support Vector Machine (SVM) • GRU (Deep Learning)	• Count Vectorizer (Bag-of-Words for SVM) • Padded Integer Sequences (for GRU)
G. Darshika Nuwangani	YouTube Dataset • URL & emoji cleaning • Non-ASCII filtering • Case normalization	• Random Forest • BERT (Transformer)	• Sublinear TF-IDF (for Random Forest) • BERT Subword Tokenizer

Note: Individual preprocessing tasks were carried out per platform, but model training for all six ML and DL architectures was performed on the unified 11,818-record dataset to ensure cross-platform generalization.

3.2 Data Flow

Below is the concise end-to-end data flow representing both training and application inference stages:



3.3 Tools/libraries used

3.5 Tools and Libraries Used

Category	Tool / Library / Platform	Key Function / Purpose in Project
Development Environment	Google Colab	Cloud-based Python notebook environment used for data preprocessing, GPU-accelerated model training, and testing.
Version Control & Collaboration	GitHub	Used for source code management, collaborative version control, and maintaining member-wise feature branches.
Programming Language	Python 3.x	Core language used across all data processing, ML/DL pipelines, and UI integration.
Data Handling & EDA	Pandas, NumPy	Dataset loading, cleaning, manipulation, and merging (SMS, Email, YouTube datasets).
NLP Preprocessing	NLTK, Regex (re)	Text normalization, stop-word removal, Porter Stemming, WordNet Lemmatization, and regex cleaning.
Feature Extraction	Scikit-Learn	Sublinear TfidfVectorizer and CountVectorizer (Bag-of-Words) implementation.
Machine Learning Models	Scikit-Learn	Building and evaluating Logistic Regression, SVM, and Random Forest models.
Deep Learning Models	TensorFlow / Keras	Tokenization, Sequence Padding, and building sequence architectures for LSTM and GRU .
Transformers & Embeddings	Hugging Face (transformers)	Utilizing BertTokenizer (input_ids, attention_masks) and fine-tuning the pre-trained BERT model.
Web Interface Deployment	Gradio	Building an interactive, user-friendly web interface directly from Google Colab for real-time spam detection and result visualization.

The complete experimental execution and model development were conducted using **Google Colab**, leveraging its cloud-based GPU capabilities for training heavy models like LSTM, GRU, and BERT. Source code and project history were systematically tracked using **GitHub**. Finally, the interactive spam detector web application was built and deployed using **Gradio**, enabling real-time text evaluation, confidence score display, and user safety warnings directly from the notebook environment.

4. Individual Model Implementations

4.1 Member 01 Implementation (P.D.M. Satharani)

4.1.1 Machine Learning Model: Logistic Regression

- **Model Explanation:** Logistic Regression is a linear binary classification model that predicts the probability that a given text belongs to a specific class (Spam or Ham) using the sigmoid activation function.
- **Algorithm Details:** It maps input TF-IDF features to log-odds probabilities:

$$\hat{y} = \sigma(W^T X + b) = \frac{1}{1 + e^{-(W^T X + b)}}$$

- **Training Process:** Trained using Scikit-Learn on the combined 11,818 record dataset with an 80/20 train-test split. It uses cross-entropy loss optimization to converge quickly.
- **Hyperparameters Used:**
 - C (Inverse Regularization Strength): 1.0 (L2 Regularization)
 - solver: 'liblinear' / 'lbfgs'
 - max_iter: 1000
- **Model Results:**
 - **Logistic Regression Results:**
 - **Accuracy:** 95.77%
 - **Precision:** 98.42%
 - **Recall:** 87.36%
 - **F1-Score:** 92.56%

4.1.2 Deep Learning Model: LSTM (Long Short-Term Memory)

- **Model Explanation:** LSTM is a specialized Recurrent Neural Network (RNN) designed to learn long-term dependencies in sequential text data while mitigating the vanishing gradient problem.
- **Architecture:**
 1. Embedding Layer: Input dimension = 10,000, Output vector dimension = 100.
 2. Spatial Dropout1D Layer: 0.2 (to prevent overfitting).
 3. LSTM Layer: 128 hidden units with tanh activation.

4. Dense Output Layer: 1 unit with sigmoid activation for binary spam prediction.

- **Training Process:** Text sequences padded to max length = 100. Compiled using Adam optimizer and trained for 10 epochs with Early Stopping.
- **Hyperparameters Used:**
 - Optimizer: Adam (Learning rate = 0.001)
 - Loss Function: binary_crossentropy
 - Batch Size: 64
 - Epochs: 10
- **Model Results:**
 - **LSTM Results:**
 - **Accuracy:** 95.81%
 - **Precision:** 96.23%
 - **Recall:** 89.61%
 - **F1-Score:** 92.80%

4.2 Member 02 Implementation (M.K.D.S. Amarasinghe)

4.2.1 Machine Learning Model: Support Vector Machine (SVM)

- **Model Explanation:** SVM finds the optimal high-dimensional hyperplane that maximizes the margin separating Spam and Ham vectors in the feature space.
- **Algorithm Details:** Uses a linear kernel function $K(x_i, x_j) = x_i^T x_j$ to handle the high dimensionality generated by Bag-of-Words/Count Vectorization.
- **Training Process:** Count-vectorized word frequencies are fed into Scikit-Learn's SVC module with hyperparameter tuning via GridSearchCV.
- **Hyperparameters Used:**
 - kernel: 'linear'
 - C: 1.0
 - gamma: 'scale'
- **Model Results:**
 - **Accuracy:** 97.76%
 - **Precision (Spam):** 98.00%
 - **Recall (Spam):** 94.00%
 - **F1-Score (Spam):** 96.00%

4.2.2 Deep Learning Model: GRU (Gated Recurrent Unit)

- **Model Explanation:** GRU is a streamlined variant of LSTM that combines the forget and input gates into a single update gate, offering computationally efficient sequential context processing.
- **Architecture:**
 1. Embedding Layer: Input vocab size = 10,000, Vector size = 128.
 2. GRU Layer: 64 GRU units with recurrent dropout = 0.2.
 3. Dense Layer: 32 units with ReLU activation.
 4. Output Layer: 1 unit with Sigmoid activation.

- **Training Process:** Trained in Keras/TensorFlow using padded sequences over 8 epochs with validation monitoring.
- **Hyperparameters Used:**
 5. Optimizer: Adam (lr=0.001)
 6. Loss Function: binary_crossentropy
 7. Batch Size: 32
 8. Epochs: 8
- **Model Results:**
 - **Accuracy:** 87.52%
 - **Precision (Spam):** 71.65%
 - **Recall (Spam):** 96.91%
 - **F1-Score (Spam):** 82.39%

4.3 Member 03 Implementation (G. Darshika Nuwangani)

4.3.1 Machine Learning Model: Random Forest

- **Model Explanation:** Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the majority class vote, reducing overfitting risk.
- **Algorithm Details:** Combines Bagging (Bootstrap Aggregation) and random feature selection across trees to classify sublinear TF-IDF features.
- **Training Process:** Fitted using Scikit-Learn on the full TF-IDF matrix using parallel processing (n_jobs=-1).
- **Hyperparameters Used:**
 - n_estimators: 100 (Number of trees)
 - criterion: 'gini'
 - max_depth: None
 - random_state: 42
- **Model Results:**
 - **Accuracy:** 96.77%
 - **Precision (Spam):** 97.46%
 - **Recall (Spam):** 91.03%
 - **F1-Score (Spam):** 94.14%

4.3.2 Deep Learning Transformer Model: BERT (Bidirectional Encoder Representations from Transformers)

- **Model Explanation:** BERT utilizes bidirectional self-attention mechanisms to read text in both directions simultaneously, capturing deep semantic context across platform-specific slang and patterns.
- **Architecture:**
 1. **Pre-trained Backbone:** bert-base-uncased (12-layers, 768-hidden units, 12 attention heads).
 2. **Tokenizer:** BertTokenizer outputting input_ids and attention_mask tensors.
 3. **Classification Head:** Linear Dropout (0.3) followed by a Dense Binary Classification Layer.

- **Training Process:** Fine-tuned on GPU in Google Colab using Hugging Face PyTorch/TensorFlow modules for 3 epochs with AdamW optimizer.
- **Hyperparameters Used:**
 4. Pretrained Model: bert-base-uncased
 5. Max Sequence Length: 128
 6. Learning Rate: 2e-5 (AdamW)
 7. Batch Size: 16 / 32
 8. Epochs: 3
- **Model Results:**
 - **Accuracy:** 98.26%
 - **Precision (Spam):** 97.76%
 - **Recall (Spam):** 96.11%
 - **F1-Score (Spam):** 96.93%

5. Evaluation and Comparison

This section evaluates six models: Logistic Regression, SVM, Random Forest (Machine Learning) and LSTM, GRU, BERT (Deep Learning).

5.1 Metrics

- **Accuracy:** $\frac{TP+TN}{TP+TN+FP+FN}$ (Overall correctness)
- **Precision:** $\frac{TP}{TP+FP}$ (Exactness in identifying spam)
- **Recall:** $\frac{TP}{TP+FN}$ (Ability to catch all spam)
- **F1-Score:** $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$ (Balance between Precision and Recall)

5.2 Comparative Table

Model Type	Model Name	Accuracy	Precision	Recall	F1-Score
ML	Logistic Regression	0.9577	0.9842	0.8736	0.9256
ML	SVM	0.9776	0.9800	0.9400	0.9600
ML	Random Forest	0.9677	0.9746	0.9103	0.9414
DL	LSTM	0.9581	0.9623	0.8961	0.9280
DL	GRU	0.8752	0.7165	0.9691	0.8239
DL	BERT	0.9826	0.9776	0.9611	0.9693

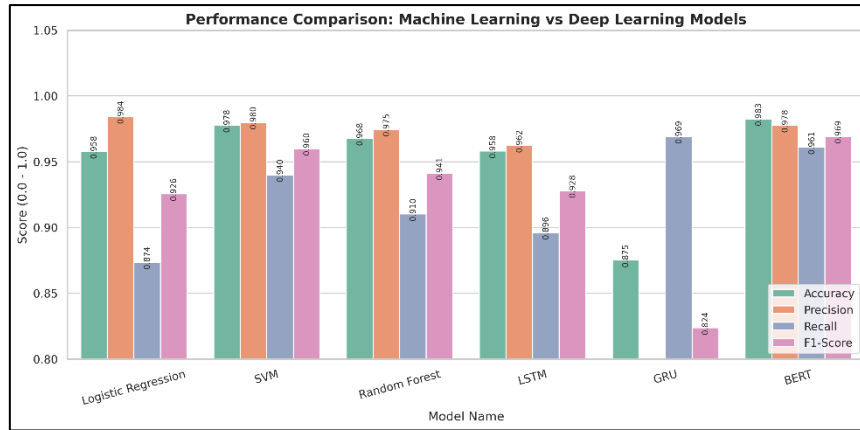


Figure 5.1: Performance Comparison: Machine Learning vs Deep Learning Models

5.3 Key Findings

- **Best Machine Learning Model:** SVM achieved the highest performance among traditional ML models (97.76% Accuracy, 96.00% F1-Score), making it the top ML choice.
- **Best Deep Learning Model:** BERT emerged as the best DL model with superior results (98.26% Accuracy, 96.93% F1-Score).
- **Other Metrics Highlights:** Logistic Regression had the highest Precision (98.42%), while GRU recorded the highest Recall (96.91%).
- **Final Model Selection:** Comparing the best ML model (SVM) and the best DL model (BERT), BERT was selected as the final overall model for this project due to its higher F1-score and ability to capture contextual nuance.

5.4 Conclusion

- **Primary Choice:** BERT is recommended as the main model for best overall accuracy and optimal performance balance.
- **Lightweight Choice:** SVM is ideal as a fast, low-resource alternative if computational power or inference speed is limited.

6. Final Application

This section outlines the implementation of the deployed Cross-Platform Spam & Threat Detector using the fine-tuned BERT model and a Gradio web interface.

6.1 System Architecture

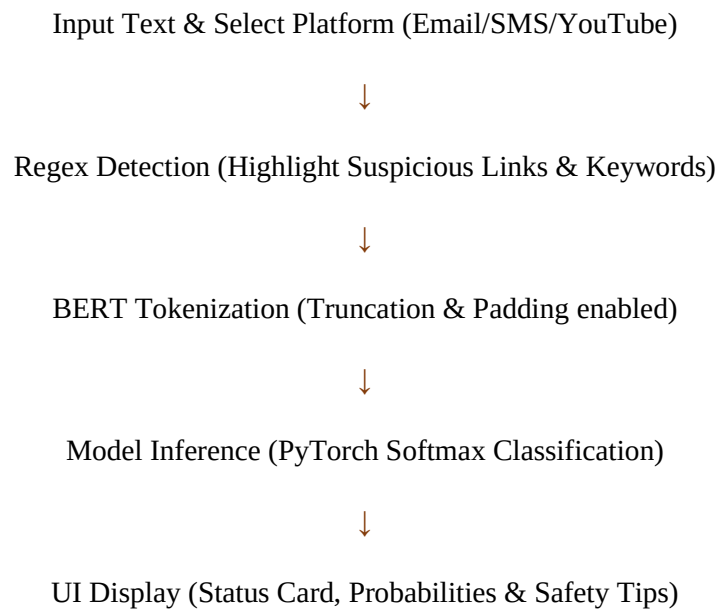
The application follows a 3-tier architecture:

- **Frontend (User Interface):** Built with Gradio for interactive input and result displays.
- **Pre-Processing Engine:** Tokenizes input via BERT Tokenizer (max_length=128) and scans for URLs/keywords using Regex.
- **Inference Backend:** Powered by PyTorch & Hugging Face Transformers to execute the BERT model and compute probabilities.

6.2 Key Features

- **Multi-Platform Support:** Tailored analysis for Email, SMS, and YouTube comments.
- **Probability Breakdown:** Displays Ham vs. Spam probabilities with a visual progress bar.
- **Smart Highlighting:** Identifies and highlights suspicious keywords and links inside the text.
- **Color-Coded Status Cards:** 🚨 Red for Spam Alerts and ✅ Green for Safe Messages.
- **Platform Security Tips:** Provides dynamic safety advice based on the selected platform.

6.3 System Workflow



6.4 Application Screenshots

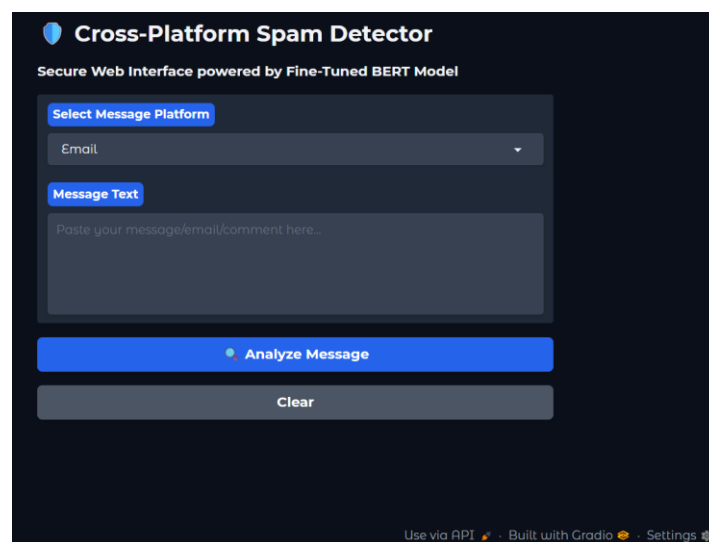


Figure 6.1: Main UI Overview

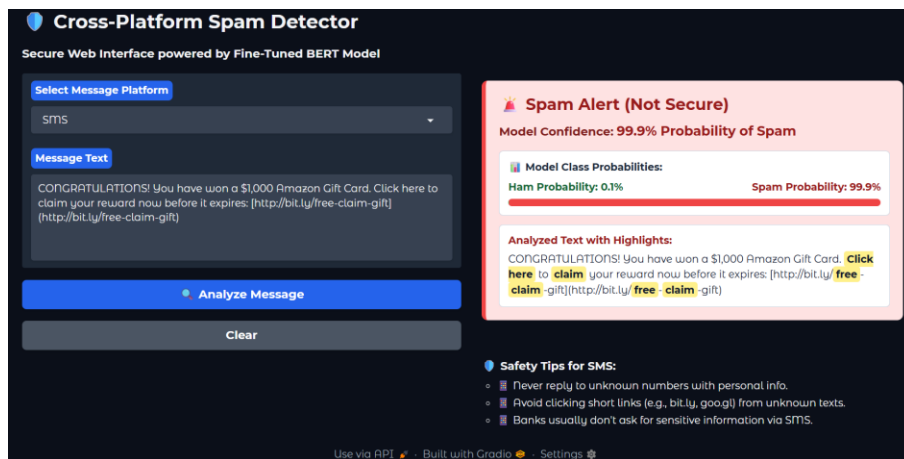


Figure 6.2: Spam Detection with Keyword Highlighting Example with Safe Message

Note: The final web application is implemented using **Gradio** and powered by the fine-tuned BERT model. Since the Gradio live interface requires active code execution in Google Colab to function, a temporary web link is generated only while the notebook session is running. Complete, step-by-step instructions on how to run the notebook and access the live application are provided in the **README.md** file of our GitHub repository.

7. Ethics and Bias Discussion

This section addresses the potential bias, ethical concerns, social implications, and mitigation strategies associated with the Cross-Platform Spam Detector.

7.1 Potential Bias

- **Dataset Imbalance:** The training dataset contains significantly more legitimate messages (Ham) than spam (e.g., SMS dataset consists of 87% Ham and 13% Spam), which can cause the model to favor predicting messages as safe.
- **Platform & Linguistic Bias:** Data collected from specific platforms (SMS, Email, YouTube) contains specific writing styles, internet slang, or emoticons. Informal language or promotional tones might be incorrectly flagged as spam.

7.2 Ethical Risks

- **User Privacy:** Using real-world SMS, email, and social media comments introduces potential privacy concerns if personal user data is not properly handled or leaked.
- **Misclassification Risks:**
 - **False Positives:** Legitimate messages (e.g., bank notifications containing login links or self-promotional YouTube comments) may be wrongly flagged as spam, leading to missed critical information.
 - **False Negatives:** New, unseen spam patterns or complex scam messages might bypass detection, exposing users to security risks.

7.3 Social Implications

- **Over-Reliance on AI:** Users might overly depend on the automated spam detector, assuming 100% accuracy, which could lower their personal cybersecurity awareness.
- **Communication Barrier:** Important professional or transactional messages being falsely marked as spam could disrupt daily business and personal communication.

7.4 Mitigation Strategies

- **Data Anonymization:** Ensure all personal identifiers are removed from datasets to protect user privacy.
- **Balanced & Diverse Training:** Combine datasets across multiple platforms (SMS, Email, YouTube) and apply data balancing techniques to reduce bias.
- **AI Confidence Transparency:** Display explicit model confidence percentages to alert users when predictions are uncertain.
- **Continuous Model Updating:** Regularly update and retrain the model with evolving spam patterns and diverse linguistic styles.
- **Decision Support Disclaimer:** Explicitly inform users that the system is a decision-support assistance tool rather than an absolute decision-maker.

8. Git Contribution Analysis

8.1 Commit History Screenshots

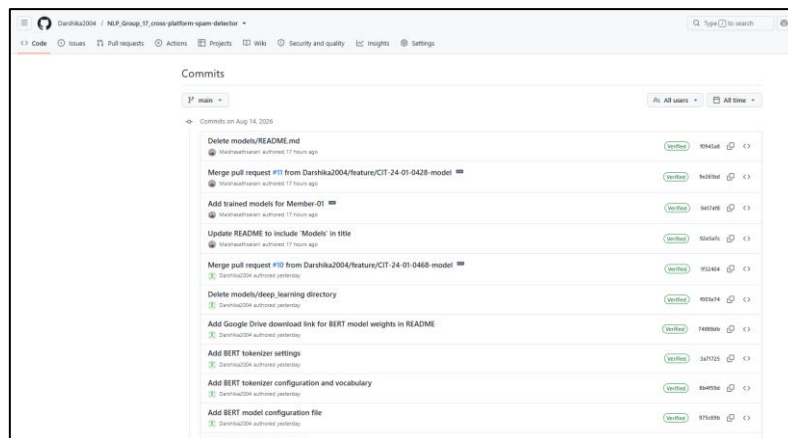


Figure 8.1: Detailed Commit Log of Group 17 Repository

8.2 Contribution Explanation

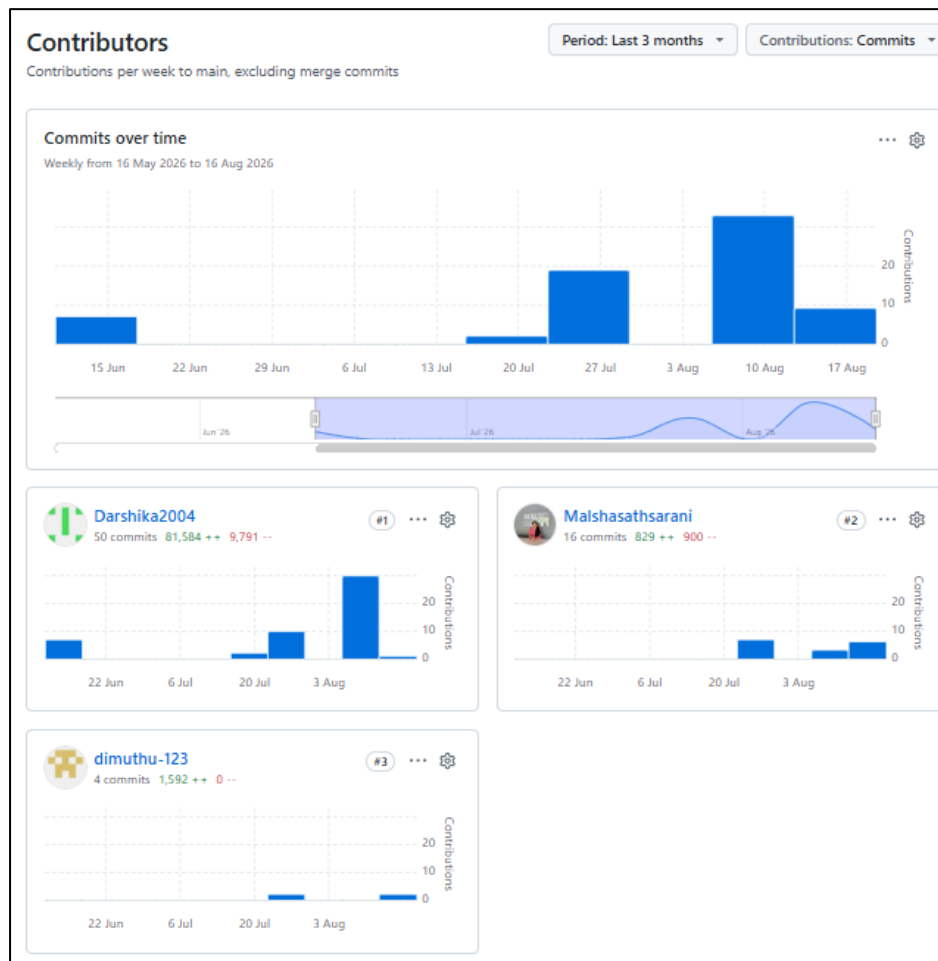


Figure 8.2: Contribution

Each group member contributed equally across dataset preparation, model building, and application integration:

- **P.D.M. Sathsarani (CIT-24-01-0428):**
 - Handled SMS dataset preprocessing and cleaning.
 - Developed and evaluated Logistic Regression and LSTM models.
 - Managed model performance metric calculations.
- **M.K.D.S. Amarasinghe (CIT-24-01-0260):**
 - Conducted Exploratory Data Analysis (EDA) and Email dataset preprocessing.
 - Developed Support Vector Machine (SVM) and GRU models.
 - Performed hyperparameter tuning and model comparison.
- **G. Darshika Nuwangani (CIT-24-01-0468):**
 - Handled YouTube dataset preprocessing and BERT Tokenization.
 - Implemented Random Forest and BERT models.
 - Developed the Gradio Web Application, final model integration, and deployment.

8.3 Branch Usage

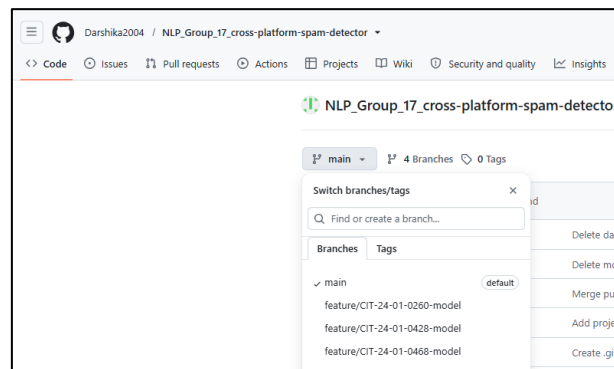


Figure 8.3: Branch Usage

To maintain clean code integration and systematic version control, a branch-per-feature workflow was utilized:

- **main:** The primary production branch containing stable code, final validated models, and the Gradio web application.
- **feature/CIT-24-01-0428-model:** Used by P.D.M. Sathsarani for SMS preprocessing, Logistic Regression, and LSTM code development.
- **feature/CIT-24-01-0260-model:** Used by M.K.D.S. Amarasinghe for EDA, SVM, and GRU model experiments.
- **feature/CIT-24-01-0468-model:** Used by G. Darshika Nuwangani for BERT implementation and Gradio app deployment.

9. Challenges and Lessons Learned

9.1 Key Challenges

- **Class Imbalance:** SMS and multi-source datasets contained significantly more Ham than Spam (e.g., 87% Ham vs 13% Spam), biasing initial model predictions.
- **Noisy Cross-Platform Text:** Handling platform-specific writing styles, mobile slang ("u", "plz"), emoticons, and obfuscated text ("m0ney", "FR3E") required extensive cleaning.
- **Resource Constraints:** Training deep learning models (BERT, LSTM, GRU) required substantial GPU memory and compute time.
- **Precision vs. Recall Trade-off:** Optimizing the model to catch spam (Recall) without falsely flagging safe messages (Precision) was difficult.

9.2 Lessons Learned

- **Text Preprocessing is Critical:** Standardizing text across SMS, Email, and YouTube is vital for model consistency.
- **Model Trade-offs:** BERT provides highest accuracy (98.26%), whereas traditional models like SVM offer faster, lightweight options (97.76%).

- **Metric Reliance:** F1-Score, Precision, and Recall are far more informative than accuracy alone for imbalanced datasets.
- **Systematic Version Control:** Feature-branch Git workflows (feature/member-name) ensured smooth team collaboration and code merging.

10. Conclusion

This project successfully developed a Cross-Platform Spam Detector by combining datasets from SMS, Email and YouTube comments. Of all the models that were evaluated, BERT has the best overall performance with an accuracy of 98.26%, demonstrating its superiority in context-aware understanding of text. The trained models were integrated into an interactive Gradio web application, allowing real-time spam detection. The web interface shows a clear confidence score so users correctly identify if a message is Spam or Ham, and it also reduces false positive risks (e.g. safe messages with links or keywords such as 'FREE'). In addition to real-time API integrations, we will also be implementing domain whitelisting, classification threshold optimisation, and continuous user-feedback retraining to further prevent such false positives in the future.

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